Manuel Fas

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SUMMARY

Machine learning researcher and engineer with experience across three research labs in Spain and Switzerland, specialising in Transformer architectures, NLP and large language models, and deep learning for sequential, tabular and biomedical data. Reimplements state-of-the-art papers into reproducible PyTorch codebases

EDUCATION

Pontifical University of Salamanca (UPSA) — **Salamanca, Spain**
Dual B.S., Computer Science and Business Administration of Technology Companies — *Sep 2022 – May 2027 (expected)*

- B.S. Business Administration completed May 2026; B.S. Computer Science in progress.
- Thesis: *Application of AI and Deep Learning to Genomic and Transcriptomic Data Analysis in Cancer* — survival modelling on TCGA RNA-seq data (819 patients, 20,440 genes); benchmarked regularised Cox models, Random Survival Forest, DeepSurv and Cox-nnet with PCA preprocessing, evaluated by concordance index and Kaplan–Meier analysis.

Lucerne University of Applied Sciences and Arts (HSLU) — **Lucerne, Switzerland**
Visiting Exchange Student, Computer Science — *Sep 2025 – Jan 2026*

EXPERIENCE

BISITE Research Group (University of Salamanca) — **Salamanca, Spain**
Junior Machine Learning Researcher — *Sep 2026 – Present*

- Research and benchmark Transformer-based architectures for financial and tabular data.
- Build reproducible PyTorch training and evaluation pipelines with leakage-safe splits, ablation studies and documented results, supporting the group's research and technology-transfer projects.
- Run local large language model inference with Ollama, profiling memory and compute bottlenecks to size hardware for research workloads.

AIR Institute — **Salamanca, Spain**
Machine Learning Research Intern — *Apr 2026 – Jun 2026*

- Investigated NLP methodologies and Transformer-based large language models to advance the SOFIA scientific initiative, surveying the literature and prototyping the most promising approaches.
- Reconstructed state-of-the-art deep learning architectures from academic papers into functional, documented PyTorch codebases, validating each reproduction against reported results.

Applied Artificial Intelligence (AAI) Lab, HSLU — **Lucerne, Switzerland**
Machine Learning Research Assistant — *Oct 2025 – Jan 2026*

- Reimplemented the TRACE paper (arXiv:2409.12972) in PyTorch, engineering Transformer encoder models over clickstream event sequences for user recommendation — github.com/Fasmanuel2002/AAI_HSLU_TRACE.
- Built Big Data preprocessing pipelines converting raw clickstream event logs into model-ready training sequences.
- Benchmarked the reproduction against published baselines and documented architecture ablations for the lab.

Wembley Studios — **Salamanca, Spain**
Machine Learning & Backend Engineer — *Jul 2025 – Sep 2025*

- Deployed machine learning models into production on Microsoft Azure, exposing them as serving endpoints for application consumption.
- Engineered high-throughput backend microservices in C# (.NET Core) integrating those ML endpoints into existing application workflows.

TECHNICAL SKILLS

Languages: Python, C#, Java, SQL
Machine Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, scikit-learn, NumPy, pandas
Engineering & Infrastructure: FastAPI, Docker, Microsoft Azure, Linux/WSL, CUDA, Ollama, Qdrant, Git, Jupyter, LaTeX
Methods: Transformer architectures, NLP & LLMs, retrieval-augmented generation (RAG), computer vision, survival analysis, paper reimplementation, experiment design and ablation studies
Spoken languages: Spanish (native), English (professional working proficiency)

CERTIFICATIONS

Generative AI with Large Language Models — DeepLearning.AI (Coursera), Aug 2026. PEFT/LoRA, RLHF (PPO, DPO).
Deep Learning Specialization (5 courses) — DeepLearning.AI (Coursera), Feb 2026. CNNs, sequence models, Transformers.
Mathematics for Machine Learning and Data Science — DeepLearning.AI (Coursera), Mar 2025.
CS50: Intro to AI with Python & Intro to Programming with Python — Harvard University (edX), Oct 2024.